

1 Model

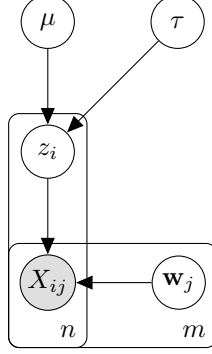


Figure 1: Graphical model without explicit deterministic feature mapping.

Let $\mathbf{z} = \{z_1, \dots, z_n\}$ be a vector of independent and identically distributed latent variables,

$$p(\mathbf{z}) = \prod_{i=1}^n \mathcal{N}(z_i \mid \mu, \tau^{-1}). \quad (1)$$

Let $\{\phi_1(\cdot), \dots, \phi_k(\cdot)\}$ denote a collection of basis functions. Define the design matrix $Z \in \mathbb{R}^{n \times k}$ implicitly by

$$Z_{i\ell} = \phi_\ell(z_i). \quad (2)$$

Let $W \in \mathbb{R}^{k \times m}$ be a weight matrix, and $X \in \mathbb{R}^{n \times m}$ denote the observed data matrix. Conditioned on $\mathbf{z}$ and W , the entries of X are independent Gaussian random variables:

$$p(X \mid \mathbf{z}, W) = \prod_{i=1}^n \prod_{j=1}^m \mathcal{N}(X_{ij} \mid (ZW)_{ij}, \beta^{-1}), \quad (3)$$

where β is a known precision parameter.

Thus, conditioned on $\mathbf{z}$, the columns of X are independent and each column $\mathbf{x}_j$ corresponds to a separate linear regression model with its own weight vector $\mathbf{w}_j$.

In order to formulate a variational treatment of this model, we assume central prior for latent variable parameters:

$$p(\mu, \tau) \propto \tau^{-1},$$

and place independent Gaussian priors on the columns of W :

$$w_j \sim \mathcal{N}(0, \alpha_j^{-1} I), \quad j = 1, \dots, m, \quad (4)$$

where $\alpha_j > 0$ is the known precision hyperparameter associated with column j .

We consider a variational distribution which factorizes between the latent variables and the parameters so that

$$q(\mathbf{z}, W, \mu, \tau) = q(\mathbf{z})q(W)q(\mu)q(\tau).$$

Let us consider the derivation of the update equation for the factor $q(W)$. Using conditional independence the log of the optimized factor is given by

$$\begin{aligned} \ln q^*(W) &= \mathbb{E}_{q(\mathbf{z}, \mu, \tau)} [\ln p(W \mid \mathbf{z}, X, \mu, \tau)] + \text{const} \\ &= \mathbb{E}_{q(\mathbf{z})} \left[\sum_j \ln p(\mathbf{w}_j \mid \mathbf{z}, X) \right] + \text{const}. \end{aligned}$$

Therefore, we have

$$\ln q^*(W) = \sum_j \ln q^*(\mathbf{w}_j),$$

where

$$\ln q^*(\mathbf{w}_j) = \mathbb{E}_{q(\mathbf{z})} [\ln p(\mathbf{w}_j \mid \mathbf{z}, X)] + \text{const}.$$

We obtain $q(\mathbf{w}_j)$ using the fact that each $\mathbf{w}_j$ is a weight vector of a bayesian linear regression model:

$$\begin{aligned} \ln p(\mathbf{w}_j \mid \mathbf{z}, X) &= -\frac{\beta}{2} \|Z\mathbf{w}_j - \mathbf{x}_j\|^2 - \frac{\alpha_j}{2} \mathbf{w}_j^\top \mathbf{w}_j + \text{const} \\ &= -\mathbf{w}_j^\top \left(\frac{\beta}{2} Z^\top Z + \frac{\alpha_j}{2} I \right) \mathbf{w}_j + \beta \mathbf{x}_j^\top Z \mathbf{w}_j + \text{const}. \end{aligned}$$

Taking the expectation with respect to $q(\mathbf{z})$, we obtain

$$\ln q^*(\mathbf{w}_j) = -\mathbf{w}_j^\top \left(\frac{\beta}{2} \mathbb{E}_{q(\mathbf{z})} [Z^\top Z] + \frac{\alpha_j}{2} I \right) \mathbf{w}_j + \beta \mathbf{x}_j^\top \mathbb{E}_{q(\mathbf{z})} [Z] \mathbf{w}_j + \text{const}.$$

The expression above is quadratic in $\mathbf{w}_j$. Since the logarithm of a multivariate Gaussian distribution is a quadratic function of its argument, it follows that

$$q^*(\mathbf{w}_j) = \mathcal{N}(\mathbf{w}_j \mid \mu_j, \Sigma_j),$$

where

$$\begin{aligned} \Sigma_j^{-1} &= \beta \mathbb{E}_{q(\mathbf{z})} [Z^\top Z] + \alpha_j I, \\ \mu_j &= \beta \Sigma_j \mathbb{E}_{q(\mathbf{z})} [Z]^\top \mathbf{x}_j. \end{aligned}$$

$$\begin{aligned}
\ln q^*(\mathbf{z}) &= \mathbb{E}_{q(W, \mu, \tau)} [\ln p(X \mid \mathbf{z}, W) + \ln p(\mathbf{z} \mid \mu, \tau)] + \text{const} \\
&= \mathbb{E}_{q(W)} [\ln p(X \mid \mathbf{z}, W)] + \mathbb{E}_{q(\mu, \tau)} [\ln p(\mathbf{z} \mid \mu, \tau)] + \text{const}.
\end{aligned}$$

$$\begin{aligned}
\ln p(X \mid \mathbf{z}, W) &= \sum_j \ln \mathcal{N}(\mathbf{x}_j \mid Z\mathbf{w}_j, \beta^{-1}I) \\
&= -\frac{\beta}{2} \sum_j \|Z\mathbf{w}_j - \mathbf{x}_j\|^2 + \text{const} \\
&= \beta \sum_i \left(-\frac{1}{2} \phi(z_i)^\top \sum_j \mathbf{w}_j \mathbf{w}_j^\top \phi(z_i) + \phi(z_i)^\top \sum_j X_{ij} \mathbf{w}_j \right) + \text{const}.
\end{aligned}$$

$$\begin{aligned}
\ln p(\mathbf{z} \mid \mu, \tau) &= -\frac{\tau}{2} \sum_i (z_i - \mu)^2 + \text{const} \\
&= \sum_i \left(-\frac{\tau}{2} z_i^2 + \tau \mu z_i \right) + \text{const}
\end{aligned}$$

$$\ln q^*(\mathbf{z}) = \sum_i \ln q^*(z_i)$$

$$\begin{aligned}
\ln q^*(z_i) &= \beta \phi(z_i)^\top \sum_j \left(-\frac{1}{2} \mathbb{E}_{q(\mathbf{w}_j)} [\mathbf{w}_j \mathbf{w}_j^\top] \phi(z_i) + X_{ij} \mathbb{E}_{q(\mathbf{w}_j)} [\mathbf{w}_j] \right) \\
&\quad + \left(-\frac{1}{2} \mathbb{E}_{q(\tau)} [\tau] z_i^2 + \mathbb{E}_{q(\mu, \tau)} [\tau \mu] z_i \right) + \text{const}.
\end{aligned}$$

$$\begin{aligned}
\ln q^*(\mu, \tau) &= \mathbb{E}_{q(\mathbf{z}, W)} [\ln p(\mu, \tau \mid \mathbf{z}, X, W)] + \text{const} \\
&= \mathbb{E}_{q(\mathbf{z})} [\ln p(\mu, \tau \mid \mathbf{z})] + \text{const} \\
&= -\frac{\tau}{2} \left(\sum_i \mathbb{E}_{q(z_i)} [z_i^2] - 2\mu \sum_i \mathbb{E}_{q(z_i)} [z_i] + n\mu^2 \right) \\
&\quad + \left(\frac{n}{2} - 1 \right) \ln \tau + \text{const}.
\end{aligned}$$

$$\ln p(\mu, \tau \mid \mathbf{z}) = \left(\frac{n}{2} - 1 \right) \ln \tau - \frac{\tau}{2} \left(\sum_{i=1}^n z_i^2 - 2\mu \sum_{i=1}^n z_i + n\mu^2 \right) + \text{const}.$$

$$\begin{aligned}\bar{z} &= \mathbb{E}_{q(z_i)} [z_i], \\ \overline{z z} &= \mathbb{E}_{q(z_i)} [z_i^2].\end{aligned}$$

$$q^*(\mu, \tau) = \mathcal{N}\left(\mu \left| \bar{z}, \frac{1}{n\tau} \right.\right) \text{Gamma}\left(\tau \left| \frac{n-1}{2}, \frac{n(\overline{z z} - \bar{z}^2)}{2} \right.\right).$$

$$\mathbb{E}_{q(\mu, \tau)} [\mu \tau] = \frac{(n-1)\bar{z}}{n(\overline{z z} - \bar{z}^2)}.$$